Accelerating toxicological research through collaboration and data reuse: The ELIXIR Toxicology Community

ELIXIR All Hands, 5-8 June 2023, Dublin, Ireland

Martens M[1], **Evelo CT**[1,2], Stierum R[3], Audouze K[4], Willighagen EL[1], ELIXIR Toxicology Community

[1] Department of Bioinformatics - BiGCaT, NUTRIM, Maastricht University, Maastricht, The Netherlands
[2] Maastricht Centre for Systems Biology (MaCSBio), Maastricht University, Maastricht, The Netherlands
[3] TNO, Risk Analysis for Products in Development, Utrecht, Netherlands
[4] Université Paris Cité, SysTox group/INSERM T3S Lab (UMRS 1124), Paris, France

1. Background

Within toxicology, as in other biomedical fields, the amount of data has been rising substantially. In particular, data derived from non-animal-based *in vitro* models employing high throughput cell-based screening assays, omics data, and imaging approaches have increased significantly over the years. These approaches are also of particular importance to creating data for novel materials, via safe and sustainable by design approaches, for which regulatory guidelines are not available or are just being conceived[1]. Moreover, to innovate risk assessment beyond currently established guideline studies, sharing and reuse of toxicological data between academic, regulatory and industrial stakeholders becomes increasingly important. To this end, the concept of FAIR (Findable, Accessible, Interoperable and Reusable) will help with making biological and chemical data interoperable.

Figure 1: The ELIXIR Toxicology Community has overlapping goals with other ELIXIR Communities and ELIXIR Platforms

2. The ELIXIR Toxicology Community

ELIXIR is an intergovernmental organization that brings together life science resources from across Europe. It recognizes the importance of toxicology and in 2020, the ELIXIR Toxicology Community was officially established with the publication of a Whitepaper[2]. The Community aims to align open data solutions from toxicology research with ELIXIR services and resources from ELIXIR Platforms and other Communities (*Figure 1*). In order to achieve this, the Toxicology Community has designed a roadmap (*Figure 2*) with specific aims to disseminate existing open solutions for data integration across toxicological domains, connect and stimulate cross-project collaborations, and developing missing open community standards to support common interests of advancing toxicology research.

Toxicology Community Roadmap

Steps	Aims
1 Disseminate existing open solutions	1. Integrating heterogeneous data types across toxicological exposure, biological, Adverse Outcome Pathways, chemical prioritization, hazard assessment, and risk assessment 2. Interoperable Software and Predictive Models 3. Ontology and Standards
2 Connect and stimulate cross-project collaborations	1. FAIRify existing community resources and solutions for findability and easier reuse 2. Increase FAIRness of relevant data 3. Inventorize Question-and-Answer (Q/A) platforms 4. Develop journal editorial standards for minimal reporting standards
3 Design and ideally develop missing open community standards	1. Engage with existing (ELIXIR) solutions 2. Sharing of standard operating procedures (SOPs) 3. Continued evaluation by adoption of existing ontologies for annotation of toxicology research output 4. Communicate and where possible disseminate FAIR data entry and/or study capture with matching software and/or templates to bench toxicologists

Figure 2: The Toxicology Community roadmap, roughly defined by three themes (steps). Each step is further detailed with possible work that could be done in ELIXIR activities to achieve its goals.

3. Discussion

Ultimately, the ELIXIR Toxicology Community's ambition is to advance data rich toxicological research to help address the challenges posed by risk assessment of existing and emerging chemicals/advanced materials. The Community will provide a platform for researchers, industry, and regulators to collaborate and share data, methods, solutions, standards, and expertise. An ELIXIR Implementation Study consisting of a workshop series is in development, focused on various aims of the Whitepaper[2].

Join the community and mailing list:

References

[1] Caldeira C, Farcal R, Garmendia Aguirre I et al. Safe and sustainable by design chemicals and materials - Framework for the definition of criteria and evaluation procedure for chemicals and materials, EUR 31100 EN, Publications Office of the European Union, Luxembourg, 2022, doi:10.2760/487955
[2] Martens M, Stierum R, Schymanski EL et al. ELIXIR and Toxicology: a community in development. F1000Research 2021, 10(ELIXIR):1129. doi: 10.12688/f1000research.74502.1

Contact

Marvin Martens
Department of Bioinformatics - BiGCaT, NUTRIM, Maastricht University
marvin.martens@maastrichtuniversity.nl
https://elixir-europe.org/communities/toxicology

These projects have received funding from the European Union's Horizon 2020 research and innovation programme

@ELIXIREurope /company/elixir-europe